idealista
estimating a property value

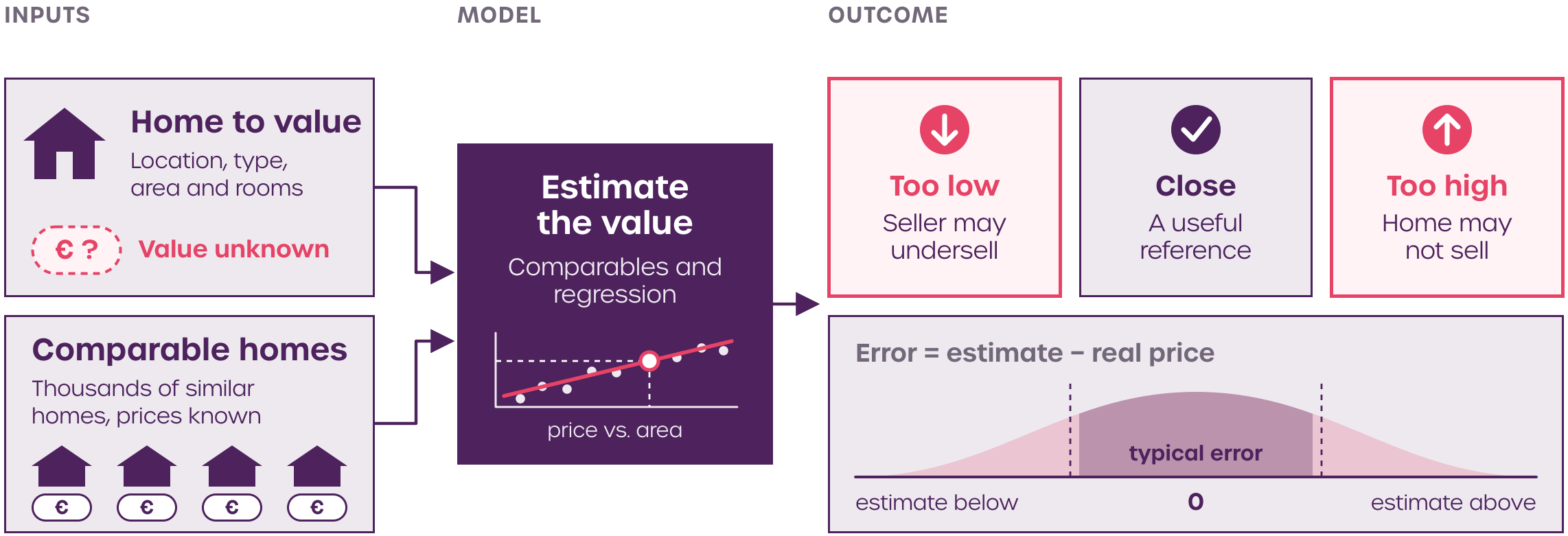
Comparables and regression, explained step by step

InnoIT · ML use case

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idealista: the problem

Valuing a home means estimating a number, and no estimate is exact



The answer is a number, not a yes or no. What matters is how far it lands from the real price.

Source: idealista/data and idealista labs

Regression analysis

idealista/data (Spain)

Commercial automated valuation service

Concept

Estimate a continuous number from explanatory features.

Advantage

Produces consistent estimates across large property portfolios.

Real use case

idealista/data combines comparables and hedonic ML to estimate property values from location and attributes.

Limitation

Missing attributes and market changes reduce accuracy. Estimates differ from completed sale prices.

The company discloses the valuation approach, but not the exact estimator.

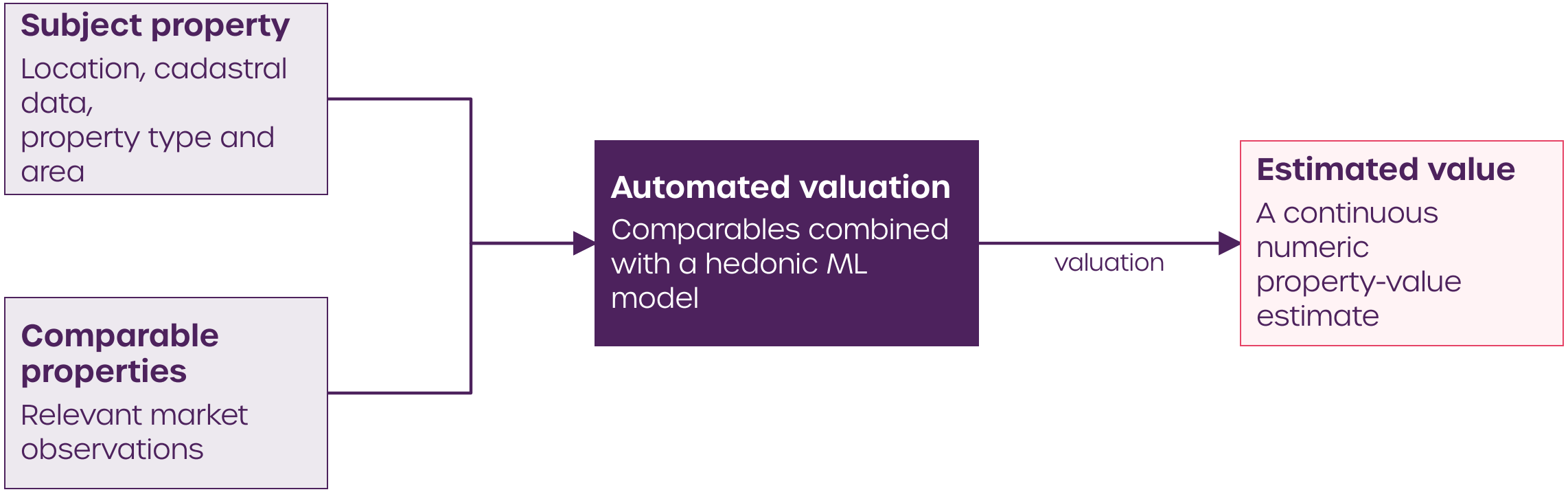
Regression estimates a number from the characteristics of an observation.

Source: idealista/data and idealista labs

idealista: estimating a property value

Property attributes and comparable properties feed automated valuation

idealista/data



Source: idealista/data and idealista labs

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1 Example | 2 Similar homes | 3 Regression | 4 Estimate | 5 Explainability

The subject property becomes a feature vector

Ana's apartment

Zone B · fictional market

Floor area	80 m²	Bedrooms	2
Renovated	Yes	Lift	Yes

↓

$$x = [80, 2, 1, 0, 1, 1]$$

area, beds, zone B, zone C, renovated, lift

Zone A is the reference. Zones B and C use separate 0/1 indicators.

Computed demo · synthetic homes and prices

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Technical details

Ana's home to value

Ana wants to estimate her apartment's price. Change its attributes to follow the same home through every step.

Floor area — 80 m²

Bedrooms — 2

Fictional zone — Zone B

Renovated — ☑

Lift — ☑

← / → steps · teaching baseline · no idealista data or proprietary model

Source: idealista/data and idealista labs

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Comparable evidence from the training set

Zone B only · 80 m² · 2 bedrooms · six closest matches

Property	Area	Beds	Renovated	Lift	Reference price
P425	85 m²	2	Yes	Yes	€302,180
P276	91 m²	2	Yes	Yes	€339,450
P201	80 m²	3	Yes	Yes	€311,444
P393	80 m²	1	Yes	Yes	€283,257
P275	85 m²	3	Yes	Yes	€311,032
P216	71 m²	2	No	Yes	€278,683

Mean of these six prices **€304,341**

All properties and prices are synthetic. Matching rules are for teaching.

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Technical details

Six similar reference properties

Find six homes like Ana's: the same zone, similar size and bedrooms. Their reference prices provide a first comparison.

What makes a match?

Same zone, similar area and bedrooms, plus renovation and lift.

Only training properties enter this comparison.

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1 Example | 2 Similar homes | 3 Regression | 4 Estimate | 5 Explainability

Learning first, prediction second

TRAINING · computed once

360 known examples
Features X + price y

→ **Linear regression**
Learn weights β

→ **OLS fit**
$\min \Sigma(y - \hat{y})^2$

PREDICTION · updates with your inputs

This home's features
80 m² · Zone B

→ **Fixed fitted model**
$\hat{y} = \beta_0 + \Sigma \beta_j x_j$

→ **Model price**
€305,790

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Technical details

Regression learns a numeric relationship

Learn from known homes, then predict Ana's price from her apartment's attributes.

What does it learn?

For example, +10 m² adds €19,802 to this model's prediction, holding other inputs fixed.

This is a fitted association in synthetic data.

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Source: idealista/data and idealista labs

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A transparent blend for this demo

Similar-home mean

€304,341

50% of the demo estimate

+

Regression estimate

€305,790

50% of the demo estimate

Illustrative estimated price

€305,000

Rounded to the nearest €1,000

The 50/50 blend is our teaching choice. idealista's production weighting is not public.

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Technical details

One continuous price estimate

For Ana's home, combine the similar-home mean with the regression prediction. This demo uses an illustrative 50/50 average.

Held-out mean absolute error

€10,577 across 120 synthetic homes.

A demo evaluation, not idealista's accuracy or an individual prediction interval.

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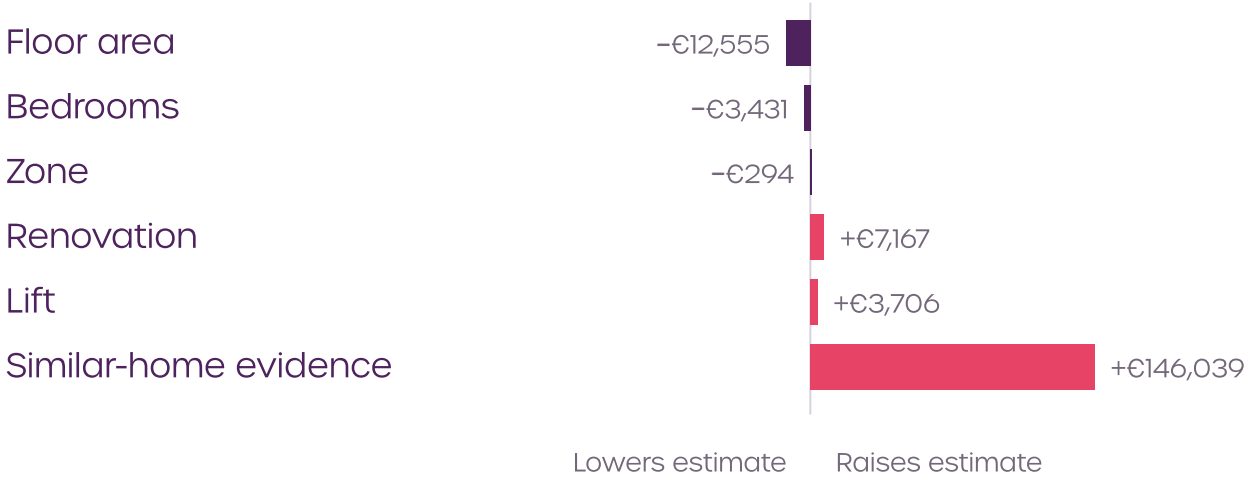
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An arithmetic breakdown of this estimate



Baseline €316,604 + adjustments **= €305,065**

Exact demo arithmetic, before rounding. The baseline is the mean training-model prediction.

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Technical details

What problem does this solve?

For Ana, explain how size, renovation and lift affect this estimate.

This apartment

80 m² versus a training average of 93 m²: floor area contributes −€12,555.

Renovation and lift

Renovated: +€7,167. Has a lift: +€3,706.

Bars explain this calculation. They do not prove causes or guarantee a sale price.

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Source: idealista/data and idealista labs

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Discussion

Which number in your business is still estimated by hand?

What would you need to predict it?

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